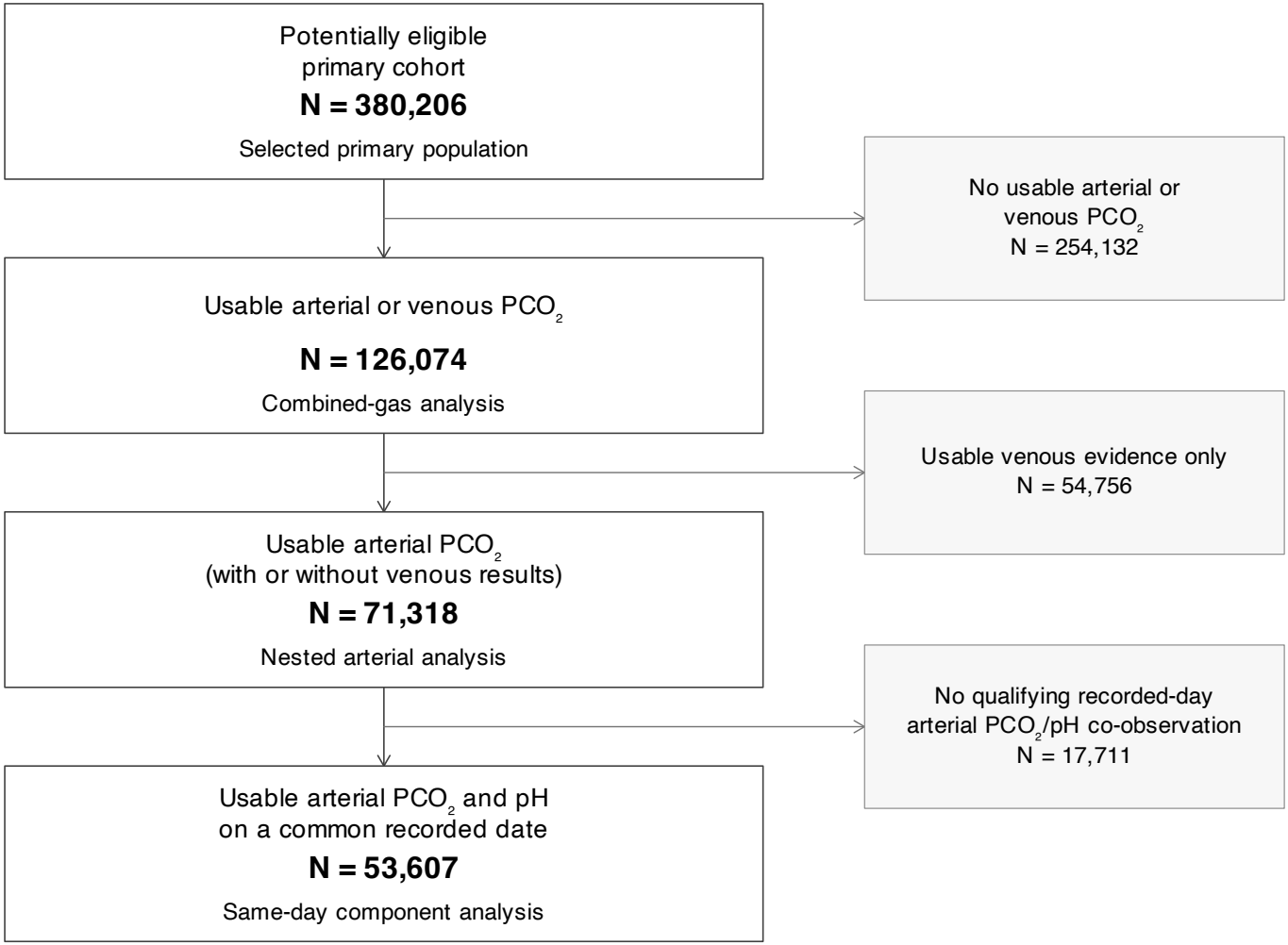
Primary gas analyses and nested arterial phenotyping

Fixed first qualifying 2022 ED/inpatient encounter; one patient per index



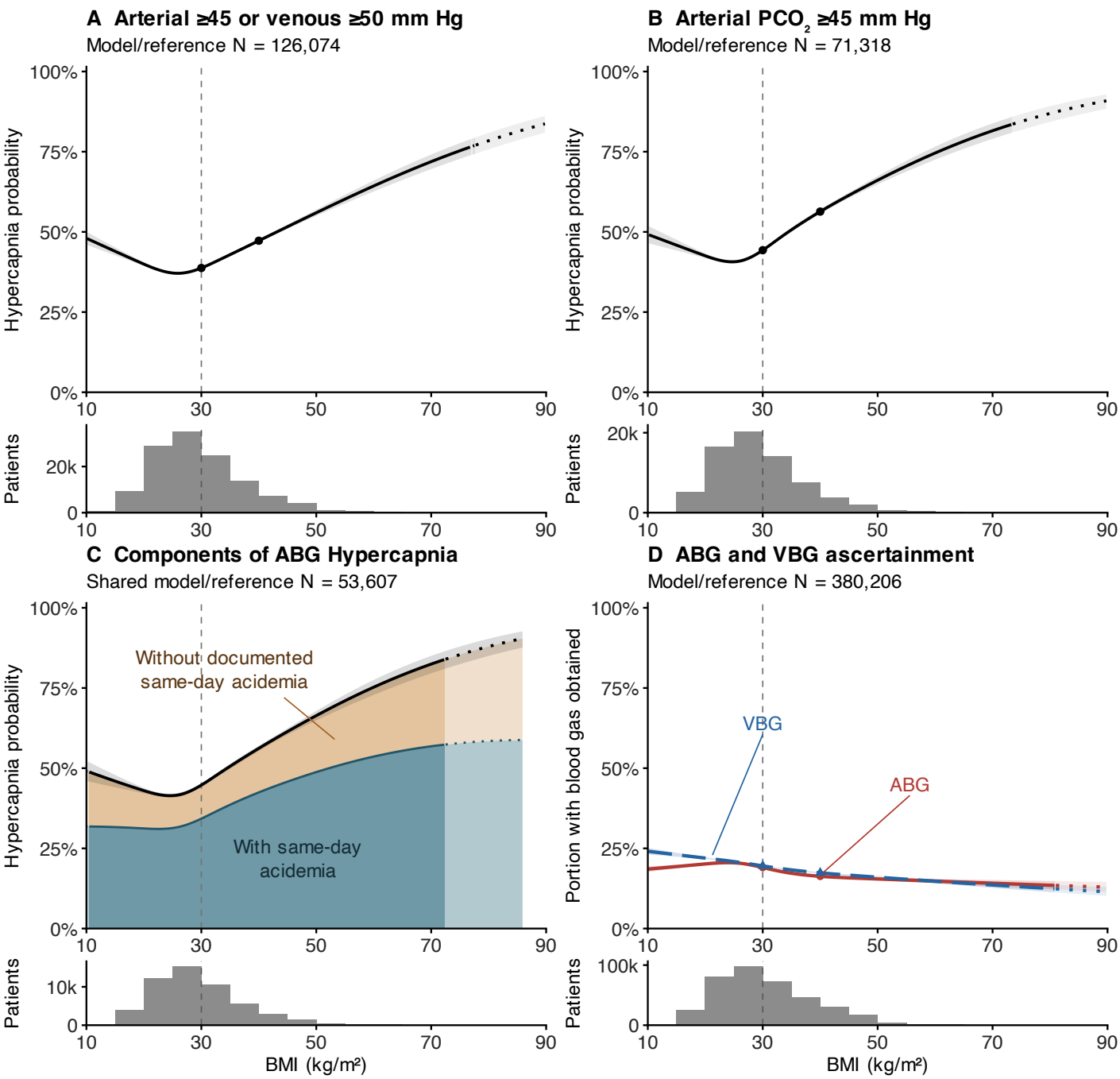
Arterial includes both-specimen patients; departures retain prior eligibility.

Same recorded-day co-occurrence; not verified same-sample pairing.

F0-primary-gas — figure caption

F0-primary-gas. Nested gas-evaluable populations at the fixed first qualifying 2022 emergency/inpatient encounter, one patient per index. The starting box is the existing primary source-entry cohort, not all TriNetX acute-care patients. Existing eligibility and index selection remain unchanged. Analysis samples retain the specified adjustment-completeness rules; any adjustment exclusions are listed separately from unavailable gas evidence. The arterial population includes patients with both arterial and venous results. Venous-only patients remain in the combined-gas analysis; patients without qualifying pH co-observation remain in the arterial analysis. No usable gas does not establish that no test was ordered or drawn. Normal and elevated PCO_2 are eligible; the last stage requires usable arterial pH, not acidemia. Usable arterial PCO_2 and pH must share at least one recorded date within that same index encounter; the date need not be its first calendar day. Same recorded-day co-occurrence; not verified same-sample pairing. Results may originate from different samples or treatment periods. Counts and evidence statuses come from the cohort ledger; unavailable recorded-day evidence is not zero.

Hypercapnia across the range of BMIs



Landmarks: standardized probability, % (pointwise 95% CI)

Outcome / evidence	BMI 30 kg/m²	BMI 40 kg/m²
Combined-gas hypercapnia	38.7% (38.4–39.1)	47.2% (46.7–47.8)
Arterial hypercapnia	44.3% (43.8–44.8)	56.3% (55.6–57.1)
With same-day acidemia	34.3% (33.7–34.8)	42.4% (41.6–43.3)
Without documented same-day acidemia	10.5% (10.1–10.8)	13.7% (13.2–14.4)
ABG ascertainment	19.0% (18.9–19.2)	16.3% (16.1–16.5)
VBG ascertainment	19.5% (19.4–19.7)	17.4% (17.2–17.6)

95% CI: gray bands (A–C total), red/blue (D); component intervals in table.
Dotted/faded: sparse evidence. Frequency strips use each panel's reference.
Same recorded-day co-occurrence; not verified same-sample pairing.

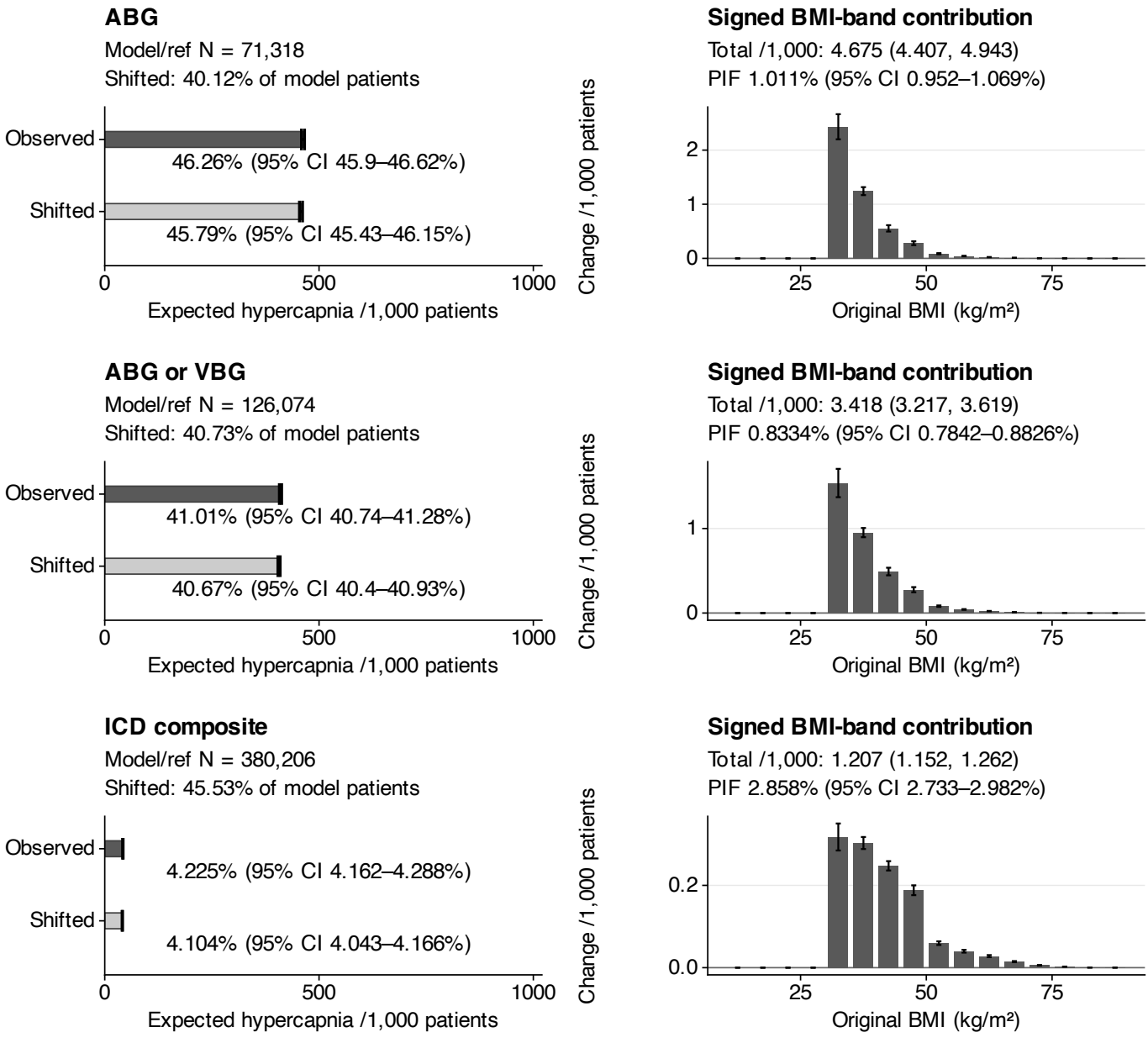
F1-main — figure caption

F1-main. Hypercapnia across the range of BMIs in the selected primary historical source-entry population at fixed first 2022 emergency/inpatient indexes. BMI: body mass index; ABG: arterial blood gas; VBG: venous blood gas; PCO_2: partial pressure of carbon dioxide; CI: confidence interval. A: arterial PCO_2 ≥45 or venous PCO_2 ≥50 mm Hg (model/reference N = 126,074). B: arterial PCO_2 ≥45 mm Hg (N = 71,318); B includes patients with both specimens; these samples overlap and must not be summed. C: joint multinomial components in patients with usable arterial PCO_2 and arterial pH on a common recorded date (N = 53,607). Blue height gives hypercapnia with documented same-day acidemia (pH <7.35); orange thickness gives hypercapnia without such a day. Any eligible day with both findings takes priority. The upper boundary and gray 95% interval describe their total, not the orange component alone. Different samples or treatment periods may contribute. Same recorded-day co-occurrence; not verified same-sample pairing. D: separate ABG (red, solid) and VBG (blue, long-dashed) models describe usable captured PCO_2 evidence across the full adjustment-complete primary reference (N = 380,206). Obtained denotes a captured, usable PCO_2 result; tests without usable captured results are not identified. Either or both specimens may be available. Normal and elevated results count, with no pH requirement, at any time within the fixed encounter. All models reuse the primary fixed three-degree-of-freedom BMI spline and adjust for age, sex, race/ethnicity, geographic region and encounter setting (EMER reference; IMP inpatient). A–C standardize over their own outcome-evaluable reference; D over all adjustment-complete primary patients. Shading and table intervals are pointwise model-based 95% confidence intervals conditional on the empirical reference and fixed basis, not intervals for a BMI 30–40 contrast. Frequency strips describe each panel’s corresponding reference. The vertical line marks BMI 30 (kg/m^2); dotted/faded segments indicate sparse or out-of-range support. Full saved curve ranges are retained. Historical selection, selective measurement, treatment timing and residual confounding limit interpretation; these results do not establish acute respiratory failure, compensation, causal effects or physiological outcomes in untested patients.

One-unit BMI reduction for observed BMI >30; no floor

Paired predictions; original reference and covariates retained

All valid BMI: primary | encounter-wide



primary/F2 — figure caption

Analysis population: All valid BMI: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F2. Paired scenario: One-unit BMI reduction for observed BMI >30; no floor. ABG: arterial PCO_2 ≥45 mm Hg; ABG or VBG: arterial ≥45 or venous ≥50 mm Hg; ICD: five-code composite including E66.2. Left: observed and shifted standardized probabilities with model-based 95% CIs and printed limits. Left axes share the full 0–1,000 reference-patient scale; printed probabilities and limits retain small differences. N identifies the common model/reference population; shifted percentage is a descriptive cohort fraction. Right: signed original-BMI-band contributions per 1,000 reference patients, with 95% CIs. Patients at BMI ≤30 remain in baseline burden and contribute exactly zero scenario difference; the eligible share is printed. Total-difference/PIF limits are printed; contribution covariance reproduces total variance. Right axes vary by outcome and retain zero and all limits. Gas samples are outcome-evaluable; ICD uses known capture. Adjustment, source-entry membership and covariates are fixed. PIF is the model-standardized potential impact fraction. These conditional scenarios do not estimate a weight-loss treatment effect.

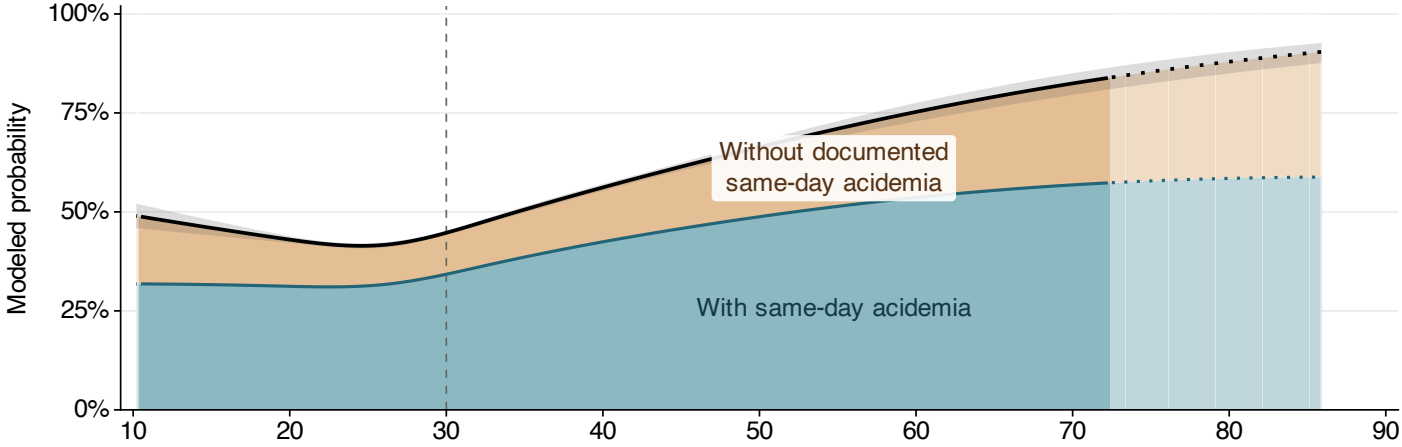
BMI and hypercapnia with and without same-day acidemia

Same recorded-day co-occurrence; not verified same-sample pairing

All valid BMI: primary; model/ref N = 53,607; with acidemia: 18,801; without: 6,200

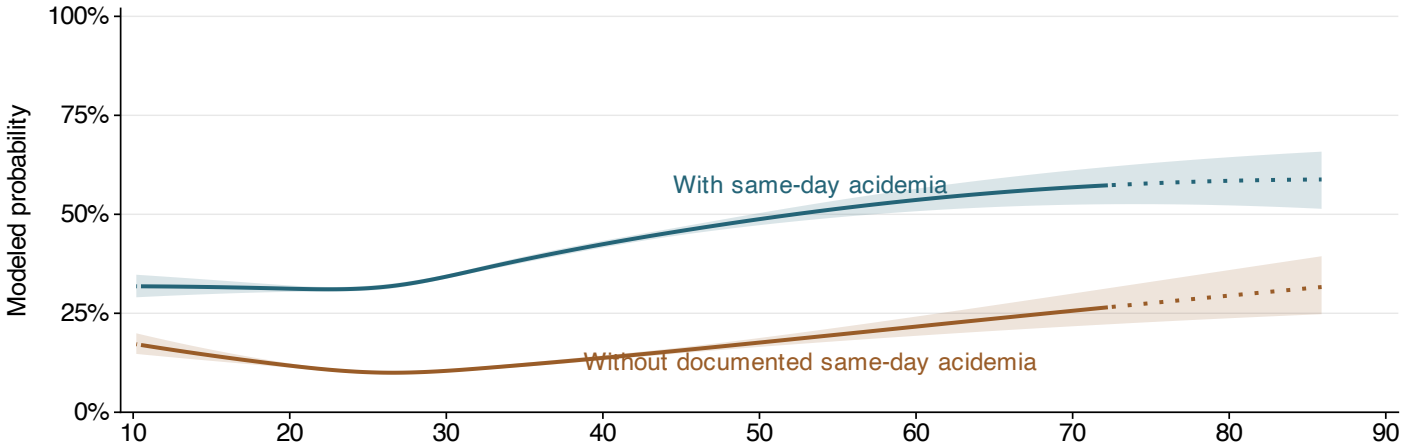
A Total hypercapnia, divided into its two components

Band thickness = component probability; top boundary = total (95% CI)

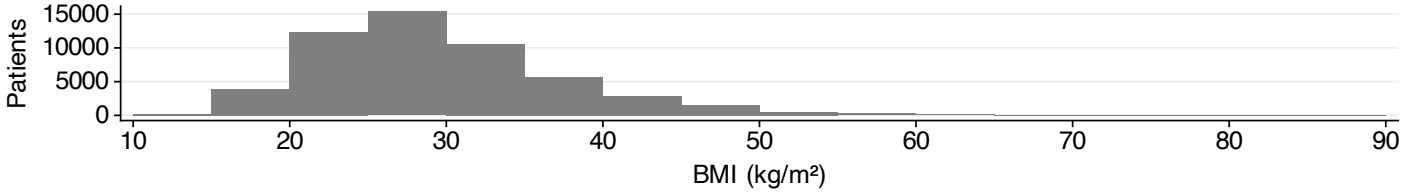


B Component curves on a common baseline

Pointwise 95% intervals; dotted segments mark sparse local evidence



C Patients in the shared co-observed population



primary/S1-same-day-ABG — figure caption

All valid BMI: primary. Same recorded-day co-occurrence; not verified same-sample pairing. Fixed-index patients with both arterial analytes share inherited plus categorical encounter-setting adjustment and their analysis population's fixed BMI spline. Any eligible day with $PaCO_2$ ≥45 mm Hg and pH <7.35 assigns the with-acidemia category; remaining hypercapnic patients have no documented same-day acidemia. Different samples may contribute. A: band thicknesses give component probabilities; the upper boundary is total hypercapnia. B: components share a baseline. Shading denotes pointwise 95% intervals using full joint covariance. C: shared-population BMI frequencies. Fading/dotted lines mark local windows below 30 patients or five events/non-events; marginal checks do not establish joint support. These recorded outcomes do not establish simultaneous physiology, acute respiratory failure or compensation. Selection, treatment timing and confounding remain limitations.

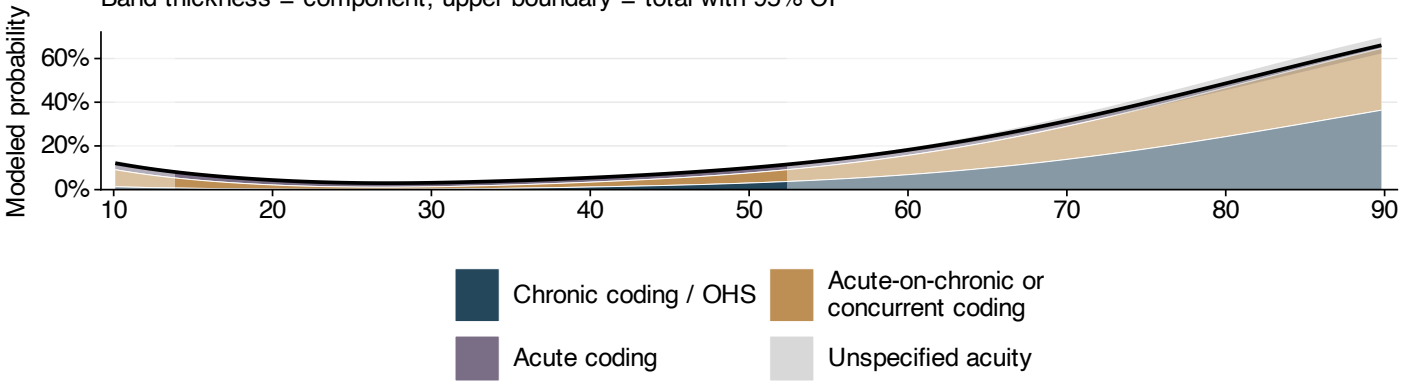
BMI and documented respiratory-failure acuity / OHS

All valid BMI: primary | Encounter-wide; model/reference N = 380,206 | Firth bias reduction

Events: chronic/OHS 2,198; acute/chronic 6,258; acute 7,339; unspecified 268

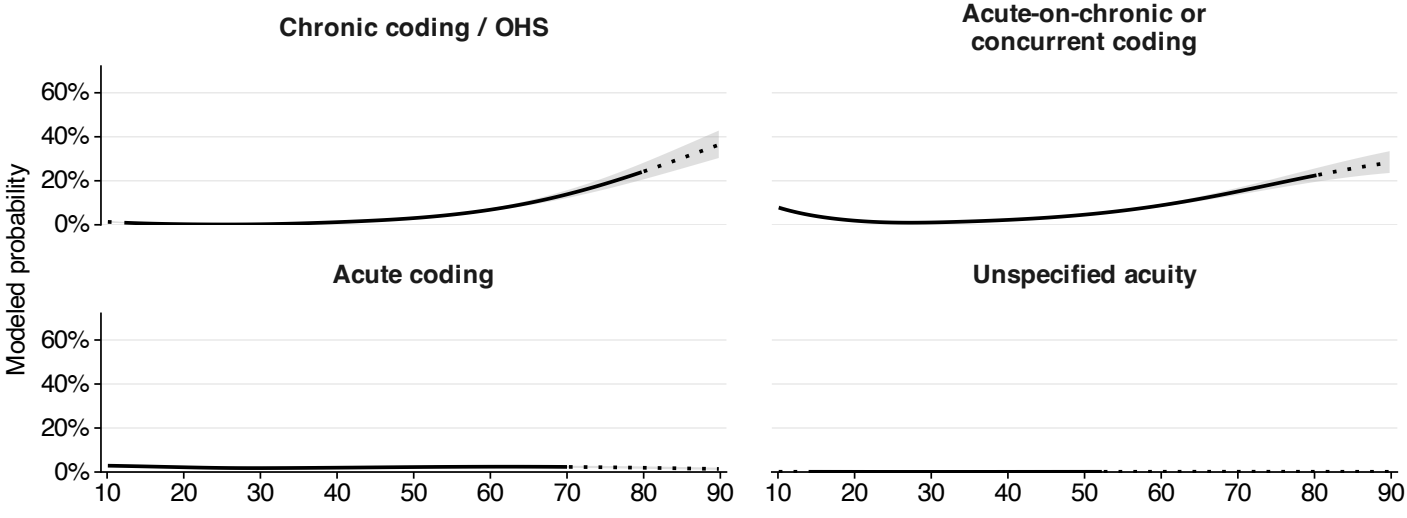
A Documented diagnoses divided by acuity coding

Band thickness = component; upper boundary = total with 95% CI

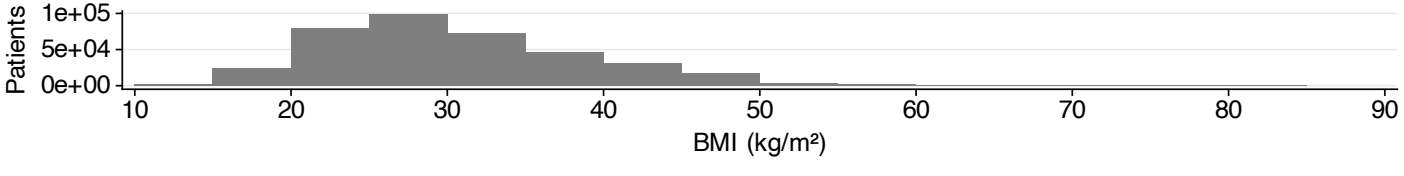


B Component magnitudes on common axes

Pointwise 95% intervals; dotted segments indicate local sparsity



C Patients in the shared diagnosis-capture population

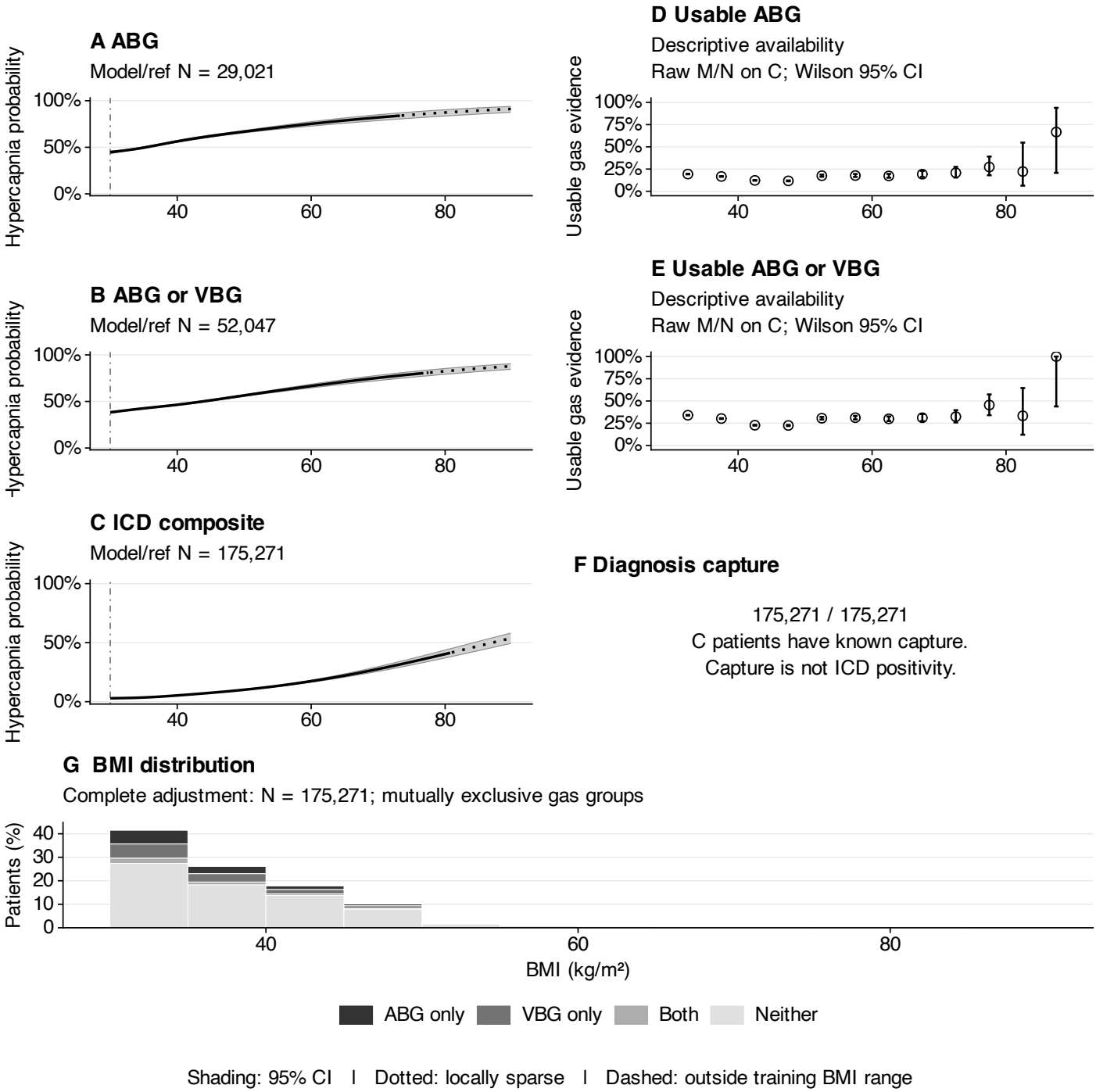


primary/S2-ICD-acuity — figure caption

All valid BMI: primary. Mean-bias-reduced five-category multinomial spline; all diagnosis-evaluable patients in the stated population retained. Encounter-wide coding, no gas requirement; inherited plus categorical encounter-setting adjustment and the population-specific spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.

BMI and documented hypercapnia

BMI ≥30: secondary | encounter-wide



Shading: 95% CI | Dotted: locally sparse | Dashed: outside training BMI range

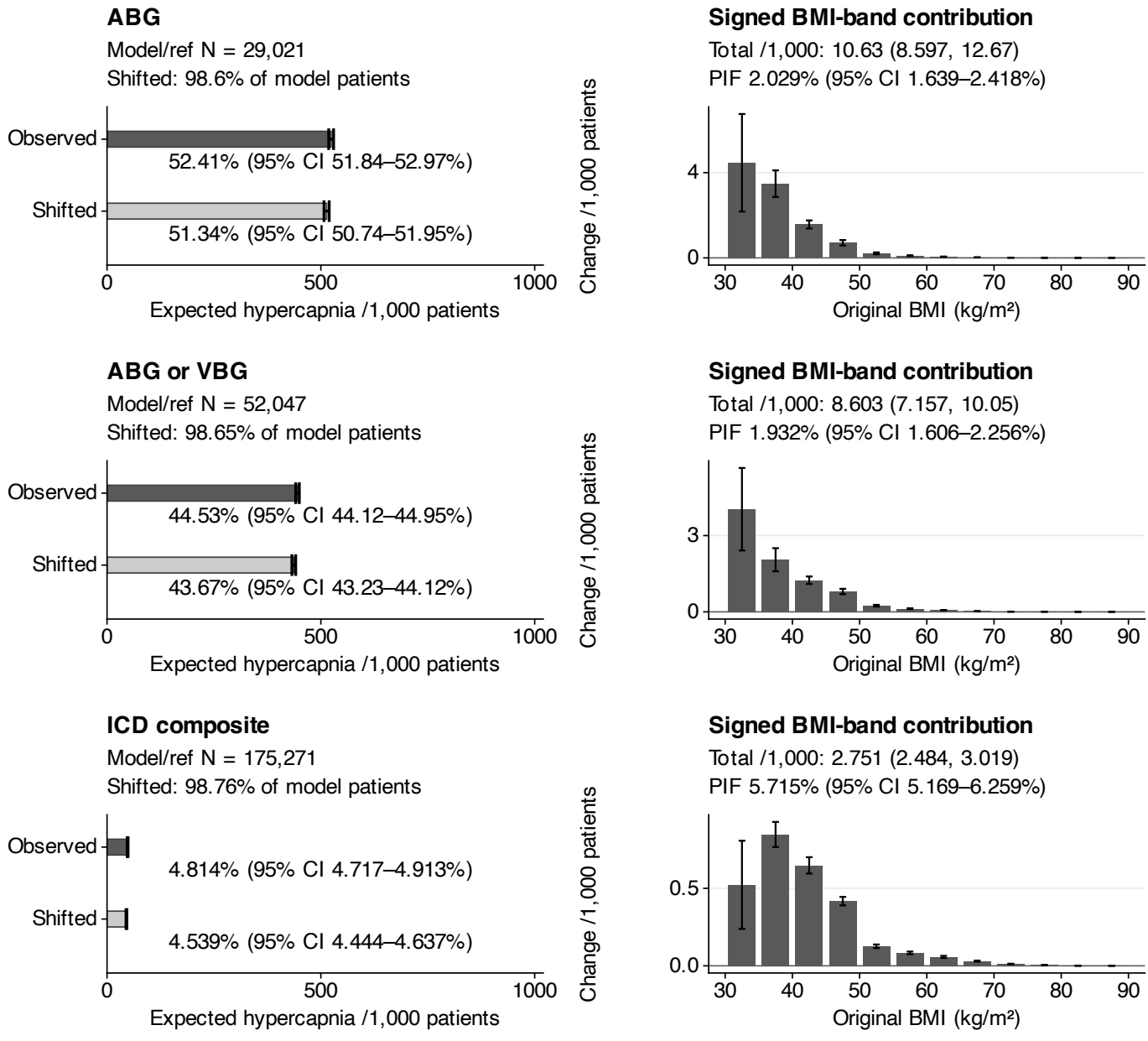
primary_bmi_ge30/F1-B — figure caption

Analysis population: BMI ≥30: secondary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. The vertical dot-dash line marks BMI 30, the scenario eligibility boundary and this secondary population's BMI cutoff. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity, geographic region and encounter setting (EMER reference; IMP inpatient) with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred.

One-unit BMI reduction for observed BMI >30; no floor

Paired predictions; original reference and covariates retained

BMI ≥30: secondary I encounter-wide



primary_bmi_ge30/F2 — figure caption

Analysis population: BMI ≥30: secondary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F2. Paired scenario: One-unit BMI reduction for observed BMI >30; no floor. ABG: arterial PCO_2 ≥45 mm Hg; ABG or VBG: arterial ≥45 or venous ≥50 mm Hg; ICD: five-code composite including E66.2. Left: observed and shifted standardized probabilities with model-based 95% CIs and printed limits. Left axes share the full 0–1,000 reference-patient scale; printed probabilities and limits retain small differences. N identifies the common model/reference population; shifted percentage is a descriptive cohort fraction. Right: signed original-BMI-band contributions per 1,000 reference patients, with 95% CIs. Patients at BMI ≤30 remain in baseline burden and contribute exactly zero scenario difference; the eligible share is printed. Total-difference/PIF limits are printed; contribution covariance reproduces total variance. Right axes vary by outcome and retain zero and all limits. Gas samples are outcome-evaluable; ICD uses known capture. Adjustment, source-entry membership and covariates are fixed. PIF is the model-standardized potential impact fraction. These conditional scenarios do not estimate a weight-loss treatment effect.

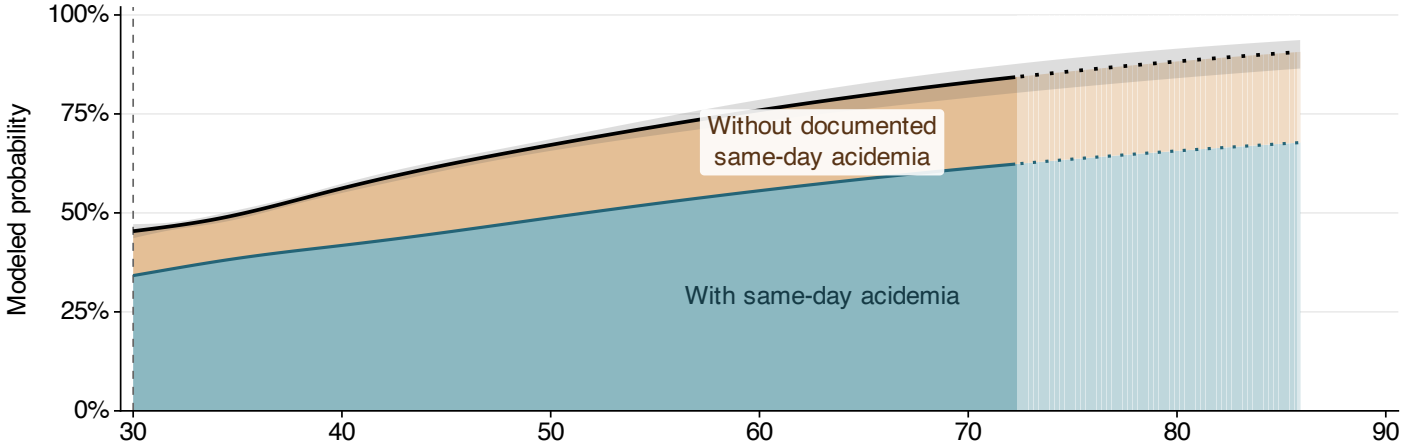
BMI and hypercapnia with and without same-day acidemia

Same recorded-day co-occurrence; not verified same-sample pairing

BMI ≥30: secondary; model/ref N = 21,748; with acidemia: 8,639; without: 2,784

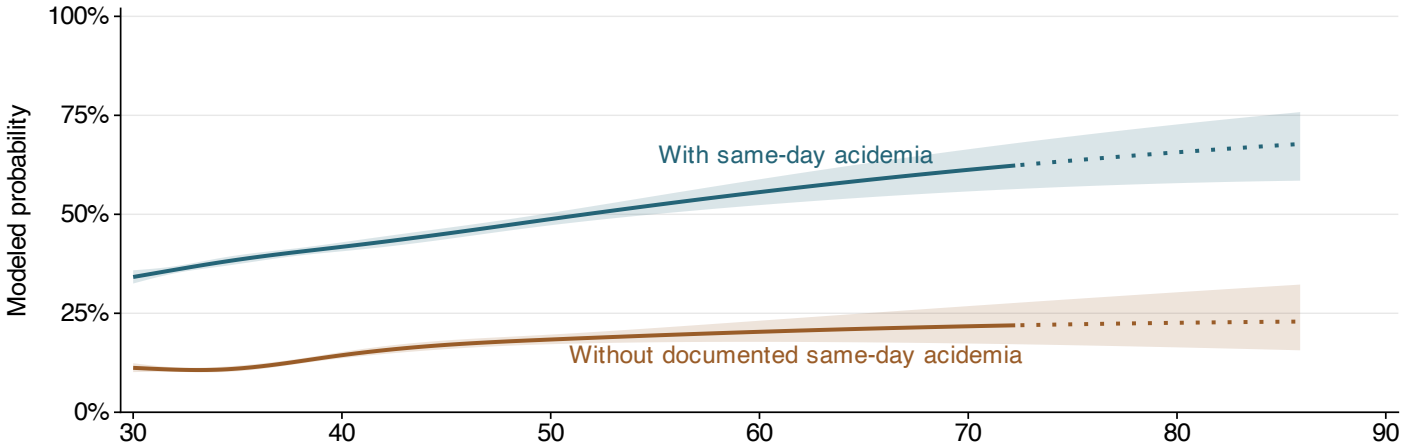
A Total hypercapnia, divided into its two components

Band thickness = component probability; top boundary = total (95% CI)

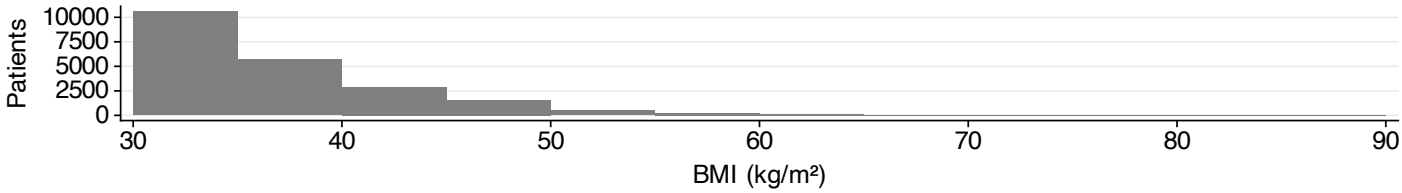


B Component curves on a common baseline

Pointwise 95% intervals; dotted segments mark sparse local evidence



C Patients in the shared co-observed population

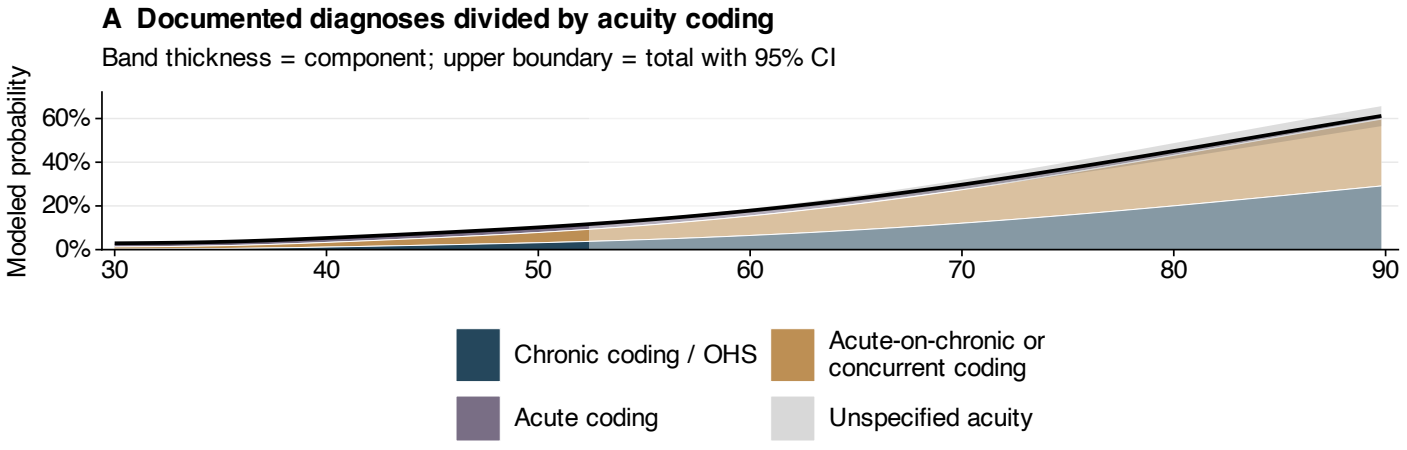


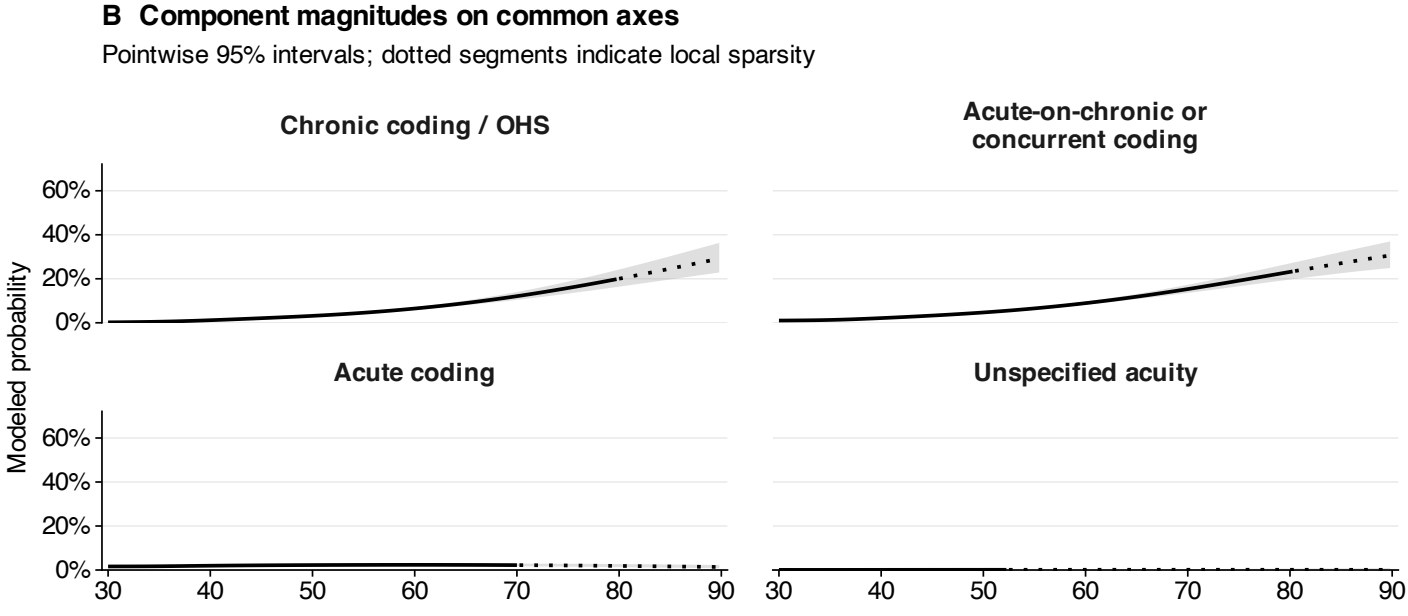
primary_bmi_ge30/S1-same-day-ABG — figure caption

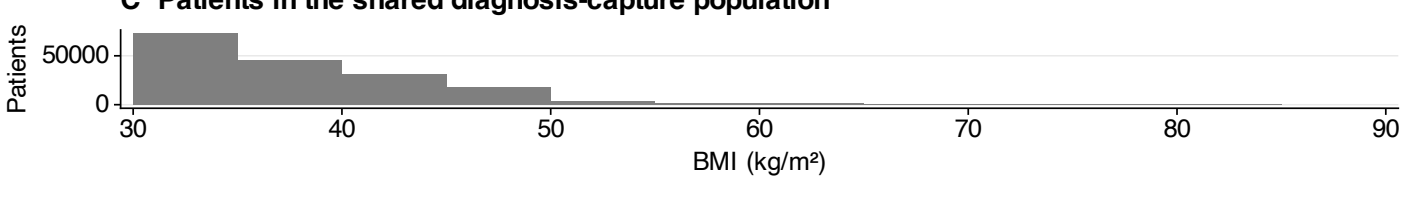
BMI ≥30: secondary. Same recorded-day co-occurrence; not verified same-sample pairing. Fixed-index patients with both arterial analytes share inherited plus categorical encounter-setting adjustment and their analysis population's fixed BMI spline. Any eligible day with $PaCO_2$ ≥45 mm Hg and pH <7.35 assigns the with-acidemia category; remaining hypercapnic patients have no documented same-day acidemia. Different samples may contribute. A: band thicknesses give component probabilities; the upper boundary is total hypercapnia. B: components share a baseline. Shading denotes pointwise 95% intervals using full joint covariance. C: shared-population BMI frequencies. Fading/dotted lines mark local windows below 30 patients or five events/non-events; marginal checks do not establish joint support. These recorded outcomes do not establish simultaneous physiology, acute respiratory failure or compensation. Selection, treatment timing and confounding remain limitations.

BMI and documented respiratory-failure acuity / OHS

BMI ≥30: secondary | Encounter-wide; model/reference N = 175,271 | Firth bias reduction
Events: chronic/OHS 1,752; acute/chronic 3,361; acute 3,201; unspecified 124







primary_bmi_ge30/S2-ICD-acuity — figure caption

BMI ≥30: secondary. Mean-bias-reduced five-category multinomial spline; all diagnosis-evaluable patients in the stated population retained. Encounter-wide coding, no gas requirement; inherited plus categorical encounter-setting adjustment and the population-specific spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.